Literature Survey

Systematic review of foundational disaster datasets, remote sensing models, and operational triage frameworks from literature.

PAPER & CITATION IN SEQUENCE	KEY FINDINGS & CONTRIBUTIONS	LIMITATIONS & UNMET CHALLENGES
[1] **Bonafilia et al. (2020)** *Sen1Floods11 • IEEE CVPRW 2020*	• Global SAR dataset (4,831 chips, 11 flood events); benchmarks all-weather U-Net flood segmentation.	■ 2D water extents only; lacks building damage labels and operational rescue task dispatch mechanism.
[2] **Gupta et al. (2019)** *Creating xBD • IEEE CVPRW 2019*	• Standardizes 850k+ buildings across 19 disasters; defines joint localization and 4-tier damage scale.	■ Optical imagery obscured by storm clouds and rain; omits flood inundation and emergency squad triage.
[3] **Rahnemoonfar et al. (2021)** *FloodNet • IEEE Access / CVPRW*	• Sub-meter post-hurricane UAV aerial imagery; combines semantic segmentation with scene VQA.	■ UAV flights limited by storm winds and battery life; lacks macro regional satellite view and Grad-CAM.
[4] **Selvakumaran et al. (2025)** *Operational USAR • Elsevier PDISAS 2025*	• Bridges satellite data with field USAR operations; uses graph belief-propagation for damage triage.	■ Lacks dynamic flood change and SAR radar modeling; no visual explainability (XAI) or automated routing.
[5] **Shakya et al. (2025)** *Pixels to Semantics • Applied AI 2025*	• Combines VRT super-resolution (4x) with YOLOv11 and Vision-Language Models (VLM) for damage rating.	■ Heavy VLM compute latency hinders field deployment; omits all-weather SAR radar and GIS task allocation.

IDENTIFIED RESEARCH GAP **The Missing Bridge: Isolated Remote Sensing vs. Actionable Rescue Dispatch**

- Workflow Silos: Existing methods treat SAR flood mapping [1], building damage [2, 5], UAV parsing [3], and USAR triage [4] in isolation.
- Operational Gap: No unified system links all-weather SAR flood masks, building damage, Grad-CAM XAI, and rescue squad dispatch in real time.